

Cisco Secure Workload + Secure Firewall

Executive Summary — Agentless Microsegmentation
2026-06-04

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Executive summary for security and network leadership

The problem

East-west traffic between application workloads is often unrestricted once an attacker is inside the network. Perimeter firewalls do not stop lateral movement. Many workloads cannot run host agents — legacy operating systems, appliances, and restricted environments — leaving gaps in segmentation coverage.

The solution

Cisco Secure Workload (CSW) and **Cisco Secure Firewall** together deliver **unified, agentless microsegmentation**: discover how applications communicate, derive least-privilege policy from observed behavior, and enforce on **Firepower Threat Defense (FTD)** devices managed by **Firewall Management Center (FMC)** — without breaking legitimate business traffic.

CSW harmonizes segmentation policy across host firewalls, network firewalls, and cloud controls into one consistent model, reducing attack surface and containing lateral movement.

How it works (two paths, one platform)

Path	What happens	Business outcome
Visibility	Firewalls export NSEL flow telemetry to CSW	See agentless workloads; map application dependencies before enforcing
Enforcement	CSW pushes L3/L4 policy to FTD via FMC	Segment east-west traffic at the network enforcement point
Workloads → Secure Firewall (FTD) —NSEL→ CSW (discover & model policy)		

└─FMC─> FTD

(enforce)

SaaS deployments use Cisco **Secure Connector** to link the cloud CSW tenant with on-premises Ingest appliances and FMC.

Three integrated security outcomes

Outcome	Description
Microsegmentation	Agentless east-west segmentation where host agents are not feasible; supports mixed agent + agentless estates
Virtual Patch	Export workload CVE intelligence to FMC; apply compensating IPS rules until permanent patches are deployed
Rapid Threat Containment	FMC detects malicious behavior → automated quarantine via CSW guardrail policies across hosts and firewalls

Deployment patterns (at a glance)

Pattern	Segmentation scope	Typical use
Layer 2 transparent firewall	Intra- + inter-subnet	Fine-grained control; legacy workloads
Layer 3 routed firewall	Inter-subnet (zone-level)	Faster time-to-segment; dev / non-prod zones
Cloud hub firewall (AWS / Azure / GCP)	Inter-VPC / inter-VNet	Centralized cloud east-west control

CSW supports **dual policy management**: CSW owns east-west microsegmentation rules; the network team retains north-south and baseline FMC policies in **Merge** mode.

Key design principles

- **Discover before enforce** — Application Dependency Mapping (ADM) and monitor mode validate policy before production impact.
- **Topology-aware enforcement** — Rules are pushed only to firewalls on the actual traffic path for each application scope.
- **Dynamic policy** — FMC **Dynamic Objects** update automatically as workloads are added, moved, or removed; no manual IP list maintenance.

- **One scope per access control policy** — Clear ownership boundary between application segmentation and firewall policy sets.
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What success looks like

Metric	Target state
Agentless visibility	Flows from firewalls visible in CSW; application dependency map validated by app owners
Policy safety	Monitor/simulate mode completed; allowed and denied test cases documented
Enforcement	Segmentation active on FTD; lateral paths blocked; business transactions pass
Operations	FMC deployment automated from CSW; compliance alerts on policy deviation

Prerequisites (leadership checklist)

- Cisco Secure Workload tenant (SaaS or on-premises)
 - Secure Workload **Ingest appliance** and **Secure Connector** (SaaS)
 - **FMC-managed FTD** devices in scope with verified deploy path
 - Application scope and ownership defined before enforcement
 - Change window and rollback plan for first enforce wave
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Official references

Resource	Link
Secure Workload & Firewall deep dive	secure.cisco.com whitepaper
FMC integration guide	Cisco.com integration guide
Step-by-step practitioner guide	GitHub — CSW-Secure-Firewall-Integration-Guide

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